

MTH 1125-Mid-Term Exam-B

Exam time: 90 minutes

Date: April 15, 2026

General instructions:

- No discussion is allowed in the exam room.
- No electronic devices, except calculators, are allowed.

PROBLEM	POINTS	SCORE
A1	5	
A2	5	
A3	5	
A4	5	
A5	5	
A6	5	
A7	5	
A8	5	
B1	10	10
B2	10	10
B3	15	12
B4	15	15
B5	10	10
TOTAL	100	97

40

PART A: Multiple choice or Short answer

(5 mark for each correct answer/choice)

Problem A1. Find the limit if it exists.

$$\lim_{x \rightarrow -\infty} (\sqrt{x^2 - 2x + 3} + x).$$

Write your answer here: 1

Problem A2. Let $f(x) = e^x + 2\sin x$, and $g(x) = x + 1$ if $x > 0$ and $g(x) = x^2 + 3x + 1$ otherwise. If $h = g \circ f$, then $h'(0) = ?$

Write your answer: $h'(0) =$ 3

Problem A3. Evaluate $\lim_{x \rightarrow -1} \frac{x^{2026} - 1}{x + 1}$.

Write your answer here: -2026

Problem A4. Find $f'(0)$ if it exists:

$$f(x) = \begin{cases} x^4 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Circle the best answer.

A. 1

B. 0

C. 2

D. It does not exist

Problem A5. Find the limit if it exists

$$\lim_{x \rightarrow -1} \frac{|x - 2| - |x + 3| + x}{x + 1}$$

Circle the best answer.

A. -1

B. $-\infty$

C. 1

D. A,B,C are incorrect

Problem A6. Find all vertical lines of the curve

$$y = \frac{2x^2 - 8x - 10}{(4x - 1)(x + 1)}$$

Write your answer: $\frac{1}{4}$

Problem A7. If $f'(x) \leq 6$ for all x and $f(2) = -1$ then the largest possible value of $f(8)$ is ..

Circle the best answer.

A. 35

B. 31

C. 50

D. A,B,C are incorrect

Problem A8. Find the 2026th derivative of $f(x) = \sin(3x)$.

Write your answer here: $-3^{2026} \cdot \sin(3x)$

PART B: WRITE YOUR FULL ANSWERS.

(If the space is not enough, please write on the back and indicate with a phrase like 'Problem ...continue on...')

Problem B1. Find all values of a and b such that f is differential everywhere:

$$f(x) = \begin{cases} a \cos x + b & \text{if } x < 0 \\ x^2 - 2x - 3 & \text{if } x \geq 0 \end{cases}$$

Problem B2. Find equations of the tangent line at the point ~~(1,0)~~ to the curve

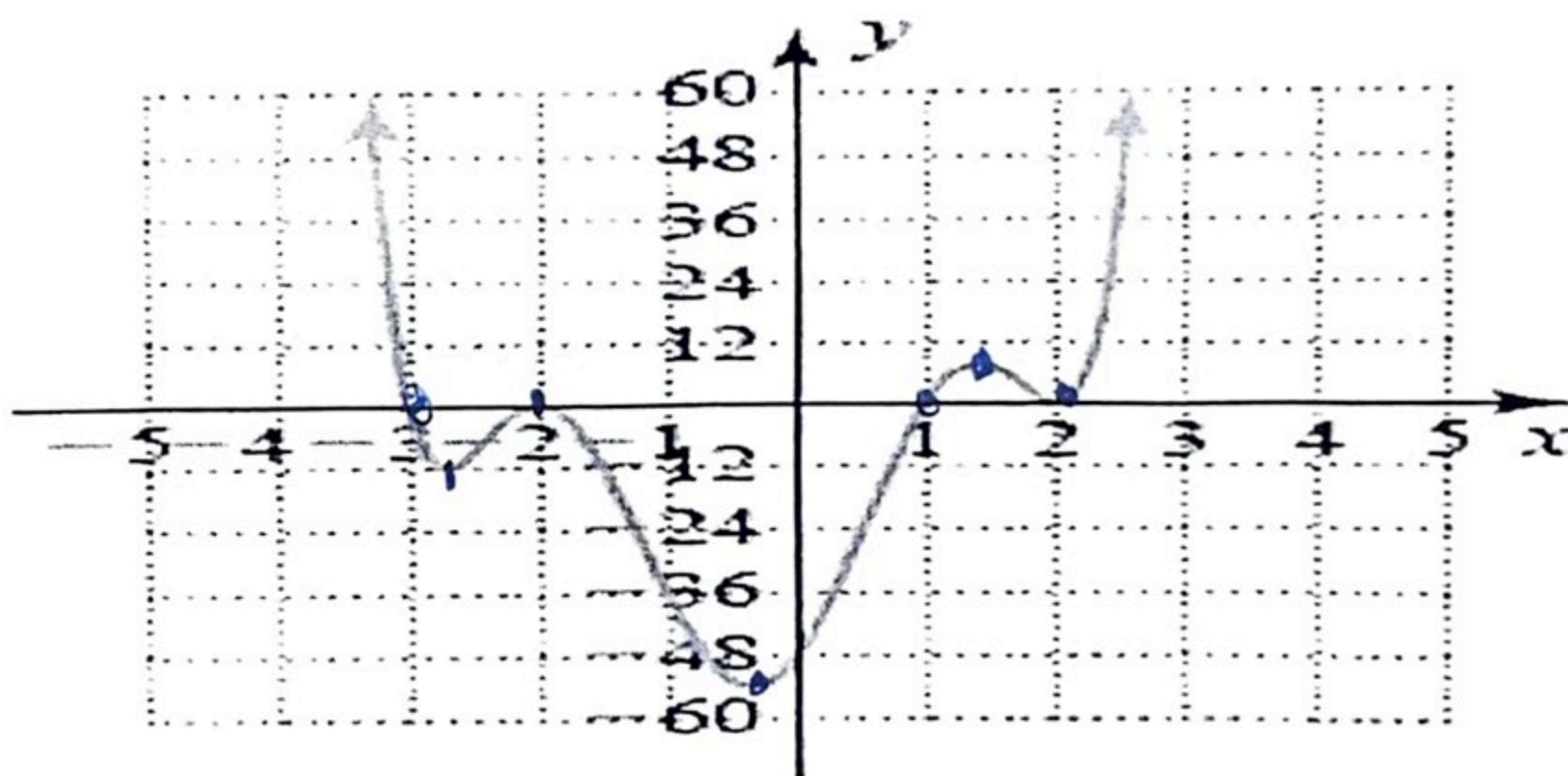
$$y = (\sin x)^{\ln x} \quad (1,1)$$

Problem B3. Let $f(x) = x^3 - 3x^2 + mx - 1$, where m is a parameter (a real number).

- Use the definition of derivative (Section 2.7, Definition 4 - Textbook) to find the derivative of f at the number a .
- Show that the equation $f(x) = 0$ has at least one real root.
- Find ~~all~~ ^{just} m such that $f(x) = 0$ has exactly one real root.

Problem B4. Given the polynomial $f(x)$ with graph below (Assume all zeroes are real).

- State $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?
- How many points where the tangent lines are horizontal? Estimate local minimal/maximal values of f ?
- Use the graph to name the zeroes of f , then state whether their multiplicity is even or odd
- state the minimum possible degree of f and write one possible function for f in factored form;



Problem B5. Suppose f is a function with the property that

$$|f(x)| \leq x^4. \quad \text{Find } f'(0)?$$

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Start to write your full solutions (PART B) below.

Problem B1:

$$f(x) = \begin{cases} a \cos x + b & \text{if } x < 0 \\ x^2 - 2x - 3 & \text{if } x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} a \cos x + b = \lim_{x \rightarrow 0^-} a \cdot \cos 0 + b = a + b$$

$$\lim_{x \rightarrow 0^+} x^2 - 2x - 3 = 0^2 - 2 \cdot 0 - 3 = -3$$

$\Rightarrow$ The function is continuous if:

$$\lim_{x \rightarrow 0^-} a \cos x + b = \lim_{x \rightarrow 0^+} x^2 - 2x - 3 \Rightarrow a + b = -3 \quad (1)$$

$$f'_{-}(0) = \lim_{h \rightarrow 0^{-}} \frac{f(0+h) - f(0)}{h}$$

$$f'_{-}(0) = \lim_{h \rightarrow 0^{-}} \frac{a \cdot \cos(h) + b - a \cdot \cos 0 - b}{h}$$

$$f'_{-}(0) = \lim_{h \rightarrow 0^{-}} \frac{a \cdot \cos(h) - a}{h}$$

$$f'_{-}(0) = \lim_{h \rightarrow 0^{-}} \frac{a \cdot (\cos(h) - 1)}{h} \quad \text{(*)} = 0 \quad \text{(**)}$$

$$f'_{-}(0) = \lim_{x \rightarrow 0^{-}} \frac{f(x) - f(0)}{x} = \frac{a \cdot (\cos x - 1)}{x} = 0 \quad \text{(***)}$$

$$\cos x - 1 \sim -\frac{x^2}{2} \Rightarrow \frac{\cos x - 1}{x} \rightarrow 0 \Rightarrow f'_{-}(0) = 0 \quad \text{(***)}$$

$$f'_{+}(0) = \lim_{h \rightarrow 0^{+}} \frac{f(0+h) - f(0)}{h}$$

$$f'_{+}(0) = \lim_{h \rightarrow 0^{+}} \frac{h^2 - 2h - 3 + 3}{h} = \lim_{h \rightarrow 0^{+}} \frac{h^2 - 2h}{h} \\ = \lim_{h \rightarrow 0^{+}} (h - 2) = -2 \quad \text{(***)}$$

$$\text{(*)}, \text{(***)} \Rightarrow f'_{-}(0) \neq f'_{+}(0) \quad (0 \neq -2).$$

$\Rightarrow$ There are no value of a and b such that f is differentiable everywhere.

Problem B2:

$$y = (\sin x)^{\ln x}$$

$$\Rightarrow \ln y = \ln (\sin x)^{\ln x}$$

$$\Rightarrow \ln y = \ln x \cdot \ln (\sin x)$$

$$\Rightarrow (\ln y)' = (\ln x \cdot \ln (\sin x))'$$

$$\Rightarrow \frac{y'}{y} = \frac{1}{x} \ln (\sin x) + \frac{\cos x}{\sin x} \cdot \ln x$$

$$\Rightarrow y' = \left(\frac{1}{x} \cdot \ln (\sin x) + \frac{\cos x}{\sin x} \cdot \ln x \right) y$$

$$\Rightarrow y' = \left(\frac{1}{x} \cdot \ln (\sin x) + \frac{\cos x}{\sin x} \cdot \ln x \right) \cdot \sin x^{\ln x}$$

$$\Rightarrow y'(1) = \ln (\sin 1)$$

Tangent line function is:

$$y = y'(x - x_0) + y_0.$$

$$y = \ln(\sin 1)(x - 1) + 1.$$

~~q~~

Problem B3:

$$f(x) = x^3 - 3x^2 + mx - 1.$$

$$a) f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{(a+h)^3 - 3(a+h)^2 + m(a+h) - 1 - a^3 + 3a^2 - ma + 1}{h}.$$

$$= \lim_{h \rightarrow 0} \frac{a^3 - 3a^2h + 3ah^2 + h^3 - 3(a^2 + 2ah + h^2) + ma + mh - 1 - a^3 + 3a^2 - ma + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3a^2 - 6a + m) + h^2(3a - 3) + h^3}{h}$$

$$= \lim_{h \rightarrow 0} 3a^2 - 6a + m + h(3a - 3) + h^2.$$

$$= 3a^2 - 6a + m.$$

$$\Rightarrow f'(x) = 3x^2 - 6x + m.$$

$$b) \text{ When } x \rightarrow +\infty \Rightarrow f(x) \rightarrow +\infty.$$

$$x \rightarrow -\infty \Rightarrow f(x) \rightarrow -\infty.$$

$$\Rightarrow \exists c \text{ in order for } f(c) = 0.$$

$$\Rightarrow f(x) = 0 \text{ has at least one real root.}$$

$$c) f'(x) = 0 \Leftrightarrow 3x^2 - 6x + m = 0.$$

$$\Delta = (-6)^2 - 4 \cdot 3 \cdot m = 36 - 12m.$$

In order to find m such that $f(x) = 0$ has exactly one real root,

$$\Rightarrow \Delta = 36 - 12m = 0.$$

$$\Rightarrow m = 3.$$

Problem B4:

a) Since the graph went up when $x \rightarrow \infty$ and $x \rightarrow -\infty$.

$$\Rightarrow \lim_{x \rightarrow \infty} f(x) = \infty \text{ and } \lim_{x \rightarrow -\infty} f(x) = \infty.$$

b) There are 5 points where the tangent line are horizontal.

local minimum / ~~maximum~~ maximum value of f is:

$$x \approx -2.8 \rightarrow f = 12.$$

$$x = -2 \rightarrow f = 0$$

$$x \approx 0 \rightarrow f = 50$$

$$x \approx 1.5 \rightarrow f = 10$$

$$x = 2 \rightarrow f = 0.$$

c) When $x = -3 \rightarrow$ odd:

$$x = -2 \rightarrow \text{even}.$$

$$~~x = -1~~ $x = 1 \rightarrow$ odd.$$

$$~~x = 0~~ $\rightarrow$ even.$$

$$x = 2 \rightarrow \text{even}.$$

All zeros are $x = -3$; $x = -2$; $x = 1$, $x = 2$.

$\Rightarrow$ The multiplicity is even.

d) The minimum possible degree of f is:

$$1 + 2 + 1 + 2 = 6.$$

$\Rightarrow$ one possible function for f is:

$$f(x) = 3 \cdot (x+3)^2 (x+2)^2 (x-1)(x-2)^2.$$

Problem B5:

$$|f(x)| \leq x^4.$$

$$|f(0)| \leq 0 \Rightarrow f(0) = 0.$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x}, \quad (*)$$

with: $|f(x)| \leq x^4$.

$$\Rightarrow \left| \frac{f(x)}{x} \right| \leq \frac{x^4}{|x|} = x^3.$$

$$x^3 \rightarrow 0 \text{ as } x \rightarrow 0.$$

$\Rightarrow$ Using Squeeze theorem.

$$\Rightarrow \frac{f(x)}{x} \rightarrow 0.$$

$$(*) \Rightarrow f'(0) = 0. \quad \checkmark$$